

Critical Path Template · Expert version

A critical path template helps you run the critical path method: list activities, durations and predecessors, calculate early and late dates and float, and find the chain of activities that sets the project's end date.

How to use this template

1. List activities, durations (days) and predecessors.
2. Forward pass: $ES = \text{the largest } EF \text{ of all predecessors}$ (0 for the first); $EF = ES + \text{duration}$.
3. Backward pass: $LF \text{ of the last activity} = \text{its } EF$; $LS = LF - \text{duration}$; work backwards using the smallest LS of successors.
4. Total float = $LS - ES$; activities with zero float are critical.
5. Focus monitoring on the critical path; see forward and backward pass.

Tips

- Re-run the calculation when durations change.
- Watch near-critical paths with little float.
- Use schedule compression only on critical activities.
- Software calculates CPM automatically for large projects.

CPM · Enter ES from the forward pass and LS from the backward pass; float and critical flag calculate

Activity	Description	Duration	Predecessors	ES	EF	LS	LF	Total float	Critical?
A	Requirements	4		0	4	0	4	0	Yes
B	Design	3	A	4	7	4	7	0	Yes
C	Procure hardware	2	A	4	6	8	10	4	No
D	Build	5	B	7	12	7	12	0	Yes
E	Install hardware	2	C	6	8	10	12	4	No
F	Test and launch	2	D, E	12	14	12	14	0	Yes